

MODULE 2

W A T E R

Conservation of Natural Resources

Course Code: BCV755B

Terra Study Materials
Department of Civil Engineering



Complete Study Notes

Sources of Water • Global & Indian Water Resources • Uses & Deficit • Inter-Basin Transfer •
Interlinking of Rivers • Groundwater • Recharge • Contamination • Seawater Ingress

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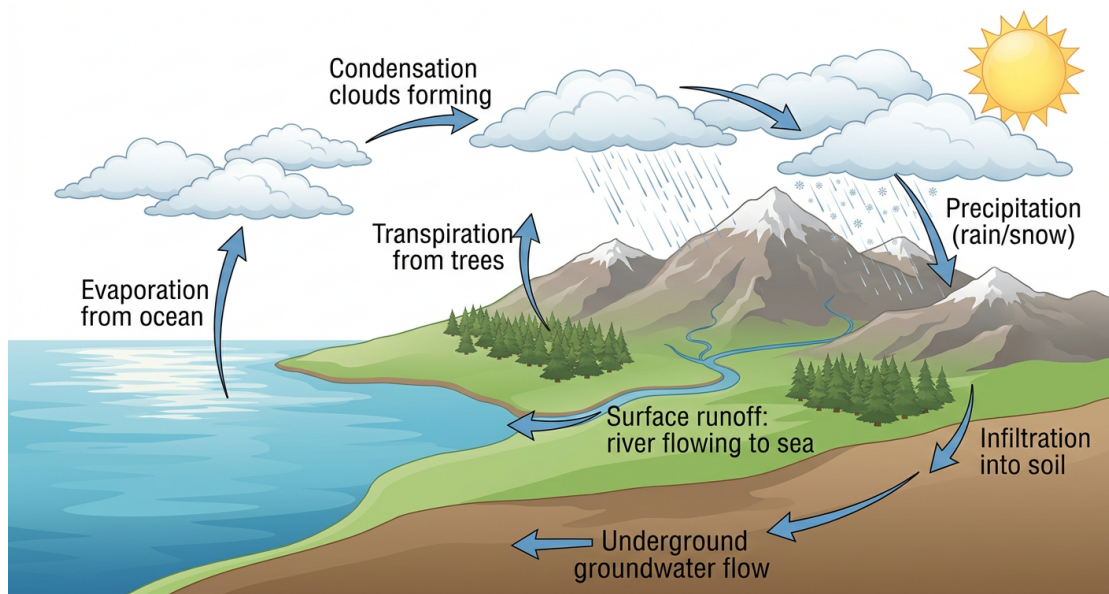
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1. Introduction to Water

1.1 Water — The Cradle of Life

- Water is an **important resource** available to all living organisms.
- It is often called the “**cradle of life**” on Earth.
- Life **started on Earth in water** and continues to survive because of the presence of water.
- Water keeps circulating through the **hydrological cycle** — evaporation, condensation, precipitation, surface runoff, underground flow, and so on.



The hydrological (water) cycle.

1.2 Two Main Sources of Water

Source	Description
Surface Water	Found in lakes, rivers, and reservoirs
Groundwater	Lies under the surface of the land; travels through and fills the openings in rocks

Key Terms

Aquifer = rocks that store and transmit groundwater (made up of gravel, sand, sandstone, or limestone). Groundwater must be **pumped from an aquifer** up to the Earth's surface before it can be used.

2. Sources of Water

There are **four main sources** from which we obtain water.

2.1 Rain Water

2.1.1 a. From roofs of houses and dwellings

- Collected and stored in **underground tanks or cisterns**.
- Used for **individual supply**.

2.1.2 b. From prepared catchments

- A catchment is given a **suitable lining and slope**.
- Water is collected in **small reservoirs**.
- Used for **community supply**.

2.2 Surface Water

- **From rivers by continuous draft** — direct extraction from flowing rivers without any diversion works.
- **From river diversions** — water is routed through a **canal into a treatment plant**.
- **From reservoir storage** — water is stored in large dams or reservoirs and then distributed.
- **From direct intake from natural lakes** — direct withdrawal from existing natural lakes.

2.3 Ground Water

- **From natural springs** — water naturally emerging from the ground.
- **From wells and bore wells** — dug wells and deep bore wells used for groundwater extraction.
- **From infiltration galleries** — horizontal tunnels/galleries dug near rivers to collect percolated water.
- **From river-side radial collector wells** — special wells with horizontal arms radiating outward to collect groundwater near rivers.

2.4 Water Obtained from Reclamation

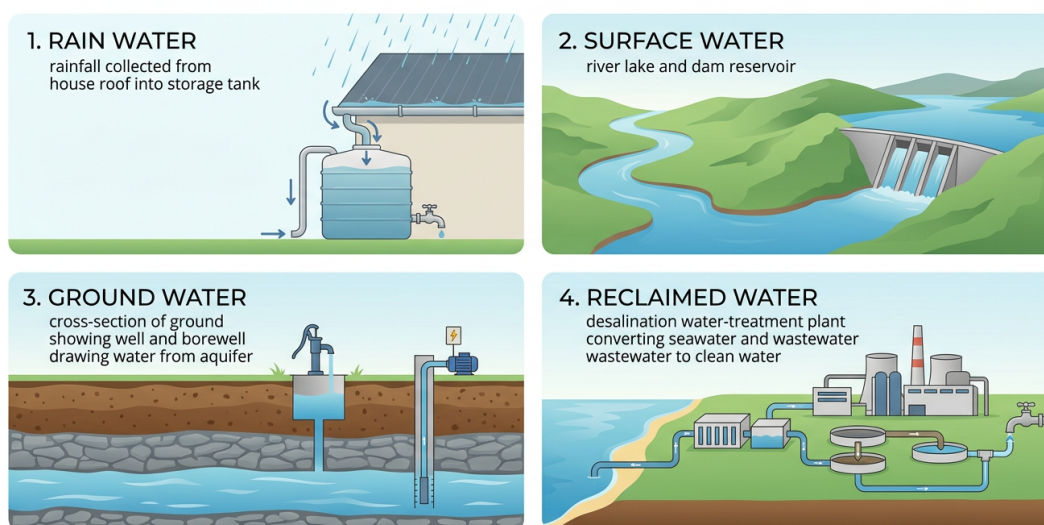
- **Desalination** — removal of salts from seawater to make it usable; used in coastal and water-scarce areas.
- **Re-use of treated wastewater** — treated sewage or industrial wastewater is reused; important for water conservation.

3. Global Water Resources

3.1 Distribution of Global Water

Type	Percentage
Saltwater (Oceans)	97.5%
Freshwater (total)	2.5%

SOURCES OF WATER



The four main sources of water.

3.1.1 Freshwater Distribution

Location	% of Freshwater
Ice caps and glaciers	79%
Groundwater	20%
Surface freshwater (lakes, rivers, soil, atmosphere)	1%

3.1.2 Surface Freshwater Breakdown

- Water in lakes: **52%**
- Water in soil: **38%**
- Water in rivers: a small %
- Water vapour in the atmosphere: a small %
- Water in living organisms: a small %

Key Point

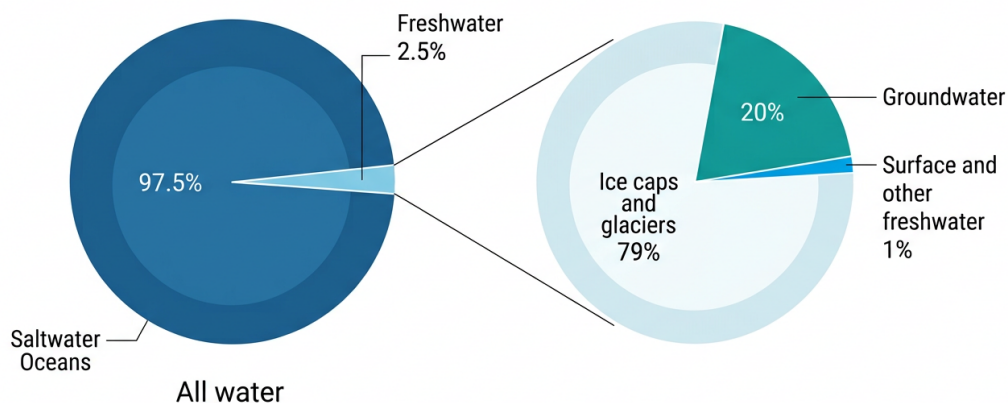
Although the Earth has an abundance of water, only a **tiny fraction** is available as freshwater for human use. This makes water a **precious and limited resource**.

4. Rivers in India — Indian Water Resources

4.1 Overview

- India is **blessed with many rivers**.
- The **land slope** determines which river the rainwater falling on an area will eventually flow into.

Distribution of Earth's Water



Distribution of the world's water.

- **River basin / catchment area** = the area from which rain will flow into a particular river.
- The **shape and size of a river basin** is determined by the topography.

4.2 Major River Basin Groups in India

4.2.1 1. Indus System

- Comprises the **River Indus** and its tributaries — **Jhelum, Chenab, Ravi, Beas, and Sutlej**.
- Originate in the **North** and flow in a **west / south-west** direction.
- Eventually flow into the **Arabian Sea through Pakistan**.

4.2.2 2. Ganga–Brahmaputra–Meghna System

- Main rivers: **Ganga, Brahmaputra, Meghna** and their tributaries.
- Ganga tributaries: **Yamuna, Sone, Gandak, Kosi**, and many others.
- All eventually flow into the **Bay of Bengal through Bangladesh**.
- Some tributaries are **larger than independent rivers** — e.g. **Yamuna** (a tributary of the Ganga) has a larger catchment area than the **Tapi** (a small peninsular river).

4.2.3 3. Rivers of Rajasthan and Gujarat

- Rivers: **Mahi, Sabarmati, Luni**, etc.
- Rivers of **arid regions** — they carry relatively **little flow**.
- Some flow to the **Arabian Sea through Gujarat**.
- Some are **land-locked** — their flow is lost through **percolation and evaporation** across the vast arid regions.

4.2.4 4. East-Flowing Peninsular Rivers

- Important members: **Damodar, Mahanadi, Brahmani, Baitrani, Subarnarekha, Krishna–Godavari, and Kaveri.**
- All flow into the **Bay of Bengal** at various places along the **eastern coast of India.**

4.2.5 5. West-Flowing Peninsular Rivers

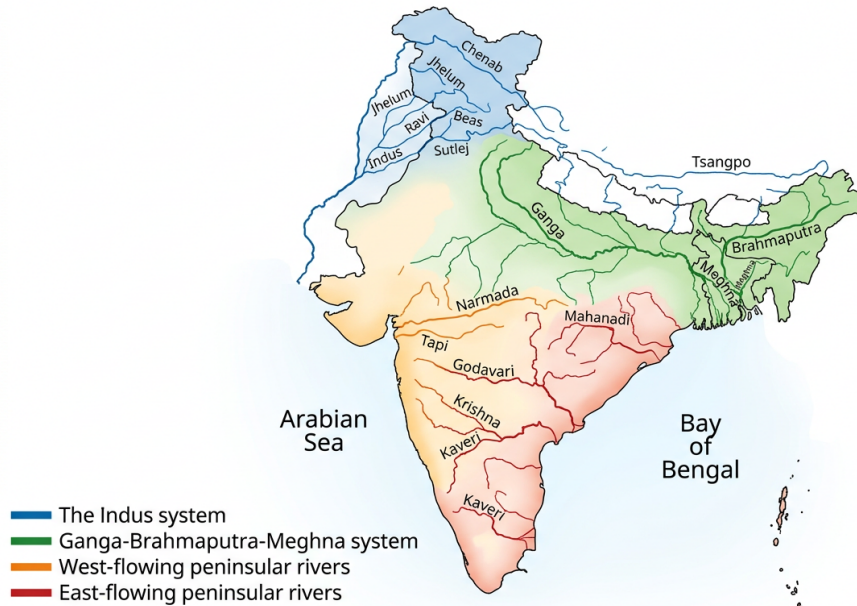
- Rivers: **Narmada and Tapi.**
- Originate in **Central India.**
- Flow in a **westward** direction to meet the **Arabian Sea** south of Gujarat.

4.2.6 6. Western Coast Rivers

- Located in **coastal Maharashtra, Karnataka, and the whole of Kerala.**
- The rivers are **small in length** but carry a **significant amount of water**, due to the **very high rainfall in the Western Ghats.**
- They drain only **3% of India's land area** but carry **11% of India's water resources.**

Key Point

This shows the **mismatch between land area and water availability** across the different regions of India.



Major river systems of India.

5. Uses of Water

5.1 Types of Water Use

Type	Description
Productive Use	Water used for irrigation of food crops, fodder crops, and medicinal herbs
Consumptive Use	Water consumed for domestic purposes — drinking, cooking, washing
Commercial Use	Water used for commercial and recreational activities

5.2 Water Use Sectors (Detailed)

5.2.1 1. Agricultural Use

- **69% of worldwide water use** is for irrigation.
- In some areas, irrigation is **necessary** to grow any crop at all.
- In other areas, it allows **more profitable crops** or **enhances crop yield**.
- It is the **single largest user of water** globally.

5.2.2 2. Industrial Use

- **15% of worldwide water use** is industrial.
- Major industrial users:
 - **Power plants** — water for cooling or as a power source (hydroelectric plants).
 - **Ore and oil refineries** — water used in chemical processes.
 - **Manufacturing plants** — water used as a solvent.
- Industrial water consumption is **generally lower** than agricultural use.

5.2.3 3. Household / Domestic Use

- **15% of worldwide water use** is for household purposes.
- Uses include **drinking water, bathing, cooking, sanitation, and gardening**.
- Basic household requirement: about **50 litres per person per day** (excluding water for gardens).

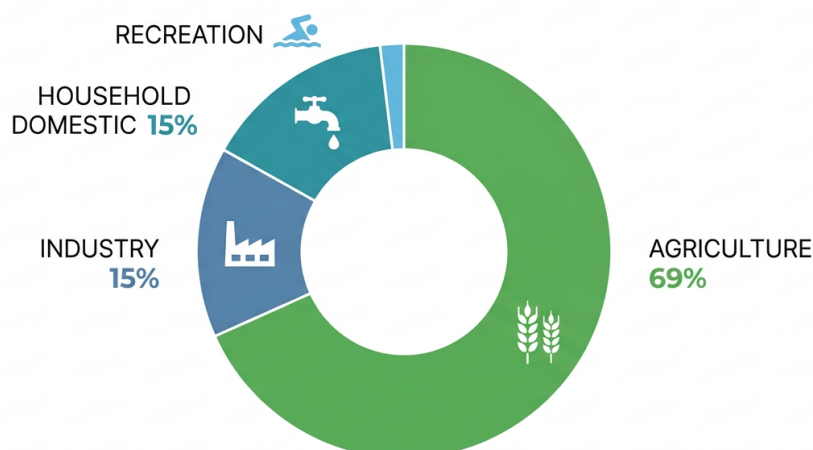
5.2.4 4. Recreational Use

- Usually a **very small but growing** percentage of total water use.
- Mostly tied to **reservoirs**.
- Examples: **anglers, water skiers, nature enthusiasts, swimmers**.
- If a reservoir is kept fuller than it otherwise would be for recreation, the water retained is categorised as **recreational usage**.

Framework for Water Allocation

The framework for allocating water resources among water users is known as “**Water Rights**”.

GLOBAL WATER USE BY SECTOR



Global water use by sector.

6. Water Deficit in India

6.1 India's Water Situation

- India has **1/6th** of the global population but only **1/25th** of the world's water resources.
- Water availability is **highly uncertain** in both **time and space**, due to India's peculiar **monsoonal climate**.
- About **70% of the population** depends — directly or indirectly — on **agriculture**.
- The urban population is **increasing rapidly**, demanding more infrastructure, amenities, and food production.
- It is essential to convert **rain-fed crops into irrigated crops**.

6.2 Drought-Prone Statistics

Parameter	Data
Drought-prone districts (Irrigation Commission, 1972)	67 districts
Number of taluks	326 taluks
States covered	8 states
Area	49.73 M Ha
CWC revised study (2006), drought-affected area	51.12 M Ha in 74 districts
Drought-prone area (% of India's total area)	About 1/6th

6.3 International Water Standards

Condition	Per-Capita Water Availability
Adequate	1700 m ³ /year
Water Stressed	1000–1700 m ³ /year
Water Scarcity	Less than 1000 m ³ /year

India's Status

India's current availability is approximately **1600 m³/year** — which places India under the **WATER STRESS** condition.

7. Inter-Basin Water Transfer (IBWT)

Inter-Basin Water Transfer (IBWT)

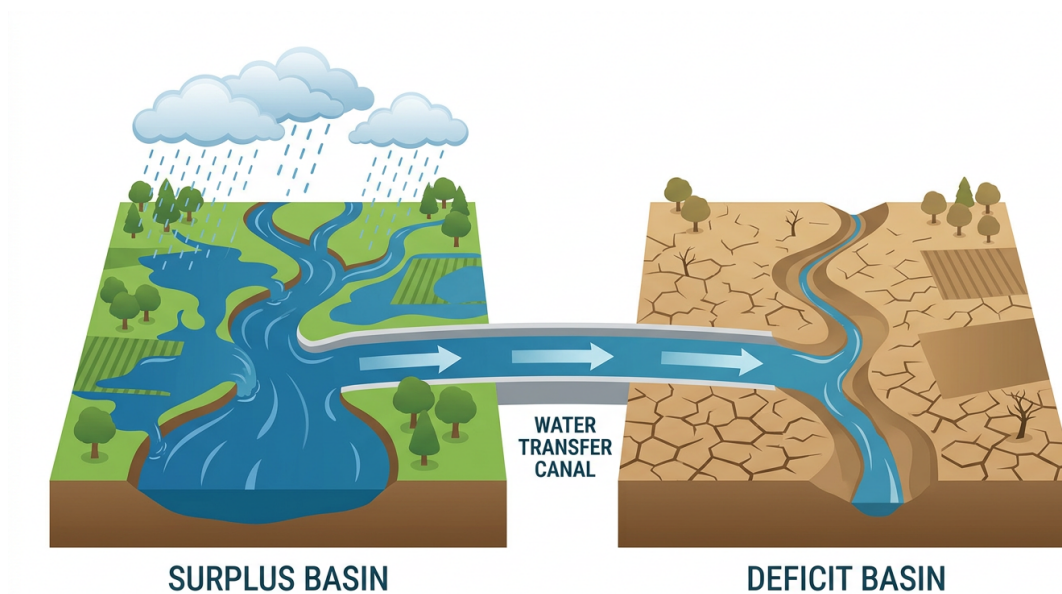
The **transfer of water from a surplus river basin to a deficit river basin**. The idea is to explore and implement IBWT in order to ensure **uniform distribution** of available natural water resources and to **mitigate the consequences** of hydrologic extremes (floods and droughts).

7.1 Why IBWT is Essential for India

- India's population is **growing at an alarming rate** — IBWT is the need of the hour.
- Helps **prevent the formation of deltas** along the coastline.
- Prevents **migration of people** from drought-affected regions.
- India's **monsoons are highly orographic**, causing **floods in the North and North-East** and **droughts in other areas**.
- The government spends **huge amounts annually** on flood and drought relief — it is better to invest in IBWT for long-term benefit.
- IBWT is seen as the **only solution** to maintain regional balance in agriculture, industry, power generation, domestic supply, and navigation.

7.2 Examples of Inter-Basin Water Transfer in India

Project	State / Year
Periyar–Vaigai project	Kerala, 1985
Kurnool–Cuddapah Canal	Andhra Pradesh, 1863–1870
Parambikulam–Aliyar project	1962–82
Telugu–Ganga project	—
Beas–Sutlej Link	1983
Indira Gandhi Nahar project (Rajasthan Canal)	1958
Sarada–Sahayak Project	1960
Ramganga–Ganga Link	1978
Tungabhadra–Pennar project	—
Mahi project	—
Tehri Multipurpose project	—



Concept of Inter-Basin Water Transfer.

8. Interlinking of Rivers

NWDA

The **National Water Development Agency (NWDA)** was **formed in 1980** to investigate and explore the possibilities of interlinking rivers.

8.1 Broad Objectives of NWDA

1. To **promote systematic and scientific study** of the interlinking of rivers.
2. To carry out **detailed study, survey, and investigation** of Peninsular and Himalayan rivers.
3. To **identify possible water-transfer projects/links** to carry water from surplus basins to deficit basins, after meeting existing and proposed needs.
4. To **prepare feasibility reports** for the various components of the Himalayan and Peninsular rivers.

8.2 NWDA's Proposal

- After thorough investigation, NWDA proposed to transfer water through a **12,500 km length of canal**.
- Inter-basin water transfer is proposed in **two components**: the **Himalayan Component** and the **Peninsular Component**.

8.3 1. Himalayan Component

8.3.1 Main Proposals

- Link the **Brahmaputra** and its tributaries with the **River Ganga**.
- Link the **Ganga** with the **Mahanadi**.
- Transfer the **surplus flows of the eastern tributaries of the Ganga to the west**.

8.3.2 Construction Envisaged

- **Canal systems and storage reservoirs** on the principal tributaries of the Ganga and Brahmaputra, in **India, Nepal, and Bhutan**.

8.3.3 Benefits

- **Additional irrigation of about 22 M Ha**.
- **Power generation of about 30,000 MW**.
- **Flood control** in the Brahmaputra and Ganga basins.

8.3.4 The 14 Links of the Himalayan Component

Link No.	Link Name
1	Brahmaputra–Ganga
2	Kosi–Ghagra
3	Gandak–Ganga
4	Ghagra–Yamuna
5	Sarada–Yamuna
6	Yamuna–Rajasthan
7	Rajasthan–Sabarmati
8	Chemar–Sone Barrage Link
9	Sone Dam–Southern Ganga
10	Ganga–Damodar–Subernarekha
11	Subernarekha–Mahanadi
12	Kosi–Mechi
13	Farakka–Sunderban
14	Brahmaputra–Ganga (Jogighopa–Tista–Farakka)

8.4 2. Peninsular Component

8.4.1 Main Proposals

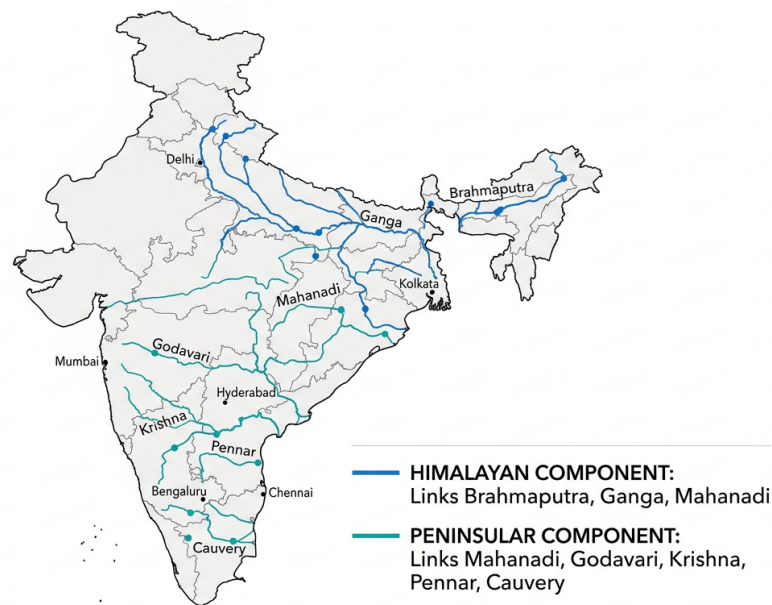
- A **network of interconnected river basins** of Peninsular India.
- NWDA carried out **water-balance studies** of all major river basins: Mahanadi, Godavari, Krishna, Pennar, Cauvery, Vaigai; the west-flowing rivers of Kerala, Karnataka, north of Bombay and south of Tapi; and the southern tributaries of the Yamuna.
- Purpose: to **find out the water-surplus and water-deficit basins**.

8.4.2 The 16 Links of the Peninsular Component

Link No.	Link Name
15	Mahanadi (Manibhadra)–Godavari (Dowleswaram) Link
16	Godavari (Inchampalli)–Krishna (Nagarjunasagar) Link
17	Godavari (Inchampalli Low Dam)–Krishna (Nagarjunasagar Tail Pond) Link
18	Godavari (Polavaram)–Krishna (Vijayawada) Link
19	Krishna (Almatti)–Pennar (Bukkapatnam) Link
20	Krishna (Srisailem)–Pennar (Mylavaram) Link
21	Krishna (Nagarjunasagar)–Pennar (Somasila) Link
22	Pennar (Somasila)–Cauvery (Grand Anicut) Link
23	Cauvery (Kattalai)–Vaigai–Gundar Link
24	Ken–Betwa Link
25	Parbathi–Kalisindh Link
26	Par–Tapi–Narmada
27	Damanganga–Pinjal Link
28	Bedti–Varda Link
29	Netravathi–Hemavathi Link
30	Pamba–Achankovil–Vaippar Link

8.4.3 Benefits of the Peninsular Component

- Possible to **irrigate about 13 M Ha**.
- About **4000 MW of hydropower** can be generated.
- Utilises **69.6 TMCum of water**.
- Involves **storage reservoirs** at potential sites, existing/ongoing reservoirs, and a network of canals for water transfer.



The two components of India's river interlinking project.

8.5 Salient Features of IBWT in India (Overall)

Feature	Data
Additional Irrigation	35 M Ha
Hydropower generation	34,000 to 40,000 MW
Cost of the project	Rs 5,60,000 crores (2002 estimate)
Water utilisation	20.96 M Ha m
Execution period	40 years
Agricultural production increase	250 to 450 Million Tons
Total length of canals	10,880 km
Cost per Ha	Rs 75,000
Water availability	1122 m ³ /capita/year
Area protected from floods	About 40 M Ha
Population protected from floods	About 260 Million
Financial savings from flood control	About 1200 crores per year
Population benefited from drought protection	About 86 Million
States benefited	14
Districts benefited	116
Financial benefit from drought protection	About 1200 crores per year
Rivers involved	37
Reservoirs involved	60

8.6 Merits of Inter-Basin Water Transfer

1. **Uniform and economic utilisation** of water resources.
2. **Enhancement** of irrigation potential and power generation.
3. Provides **ample surface water** for growing domestic and industrial needs.
4. **Scope for inland navigation** — reduces stress on the existing transport system.
5. **Minimises the intensity** of drought and floods.
6. Helps **increase per-capita income**.
7. **Reduces exploitation of groundwater** as surface water becomes abundant.
8. **Huge employment generation** during construction and operation.
9. **Development of fisheries** in canals and reservoirs.
10. **Salinity control** — fresh water helps push back saltwater ingress.
11. **Recreation facilities** from canals and reservoirs.
12. **Infrastructural development** in surrounding areas.
13. **Socio-economic development** of drought-affected regions.
14. **Improvement in groundwater potential** through artificial recharge.
15. **Control of migration** of people from drought-prone areas.
16. **Conversion of barren land** into cultivable land.
17. **Reduction in the formation of further deltas** in coastal zones.
18. **Minimisation of relief expenditure** on floods and droughts.

8.7 Demerits of Inter-Basin Water Transfer

1. **Large areas liable to submersion** due to reservoirs and canals — displacement of people.
2. **Adverse effects on the ecological system** — disruption of habitats and biodiversity.
3. **Difficult to resolve interstate / international water disputes**.
4. **Legal problems** in sharing water between states.
5. **Very high project and maintenance costs**.
6. **Water pollution in conveyance** — canals pick up pollutants along the way.
7. **Loss of water in conveyance** (seepage and evaporation).
8. **Land acquisition and rehabilitation problems**.
9. **Requires afforestation** to compensate for lost green cover.
10. **Long-term project** — large variation in estimation and planning.
11. **Serving high-altitude areas needs pumping** — huge power and maintenance costs.
12. **Needs huge public and governmental debate** at micro and macro levels.

13. **Soil erosion and sedimentation** problems in canals and reservoirs.
14. In **undulating zones**, a large number of **cross-drainage works** may be required.

9. Groundwater

Groundwater

Groundwater is water that seeps through rocks and soil and is **stored below the ground**. The rocks in which it is stored are called **aquifers**.

9.1 Explanation

- Aquifers are typically made up of **gravel, sand, sandstone, or limestone**.
- Water moves through these rocks because they have **large connected spaces** that make them **permeable**.

9.2 Key Terms in Groundwater

Term	Definition
Aquifer	Rock that stores and transmits groundwater
Saturated Zone	The area where water fills the aquifer completely
Water Table	The depth, from the surface, at which groundwater is found

9.2.1 Water Table Facts

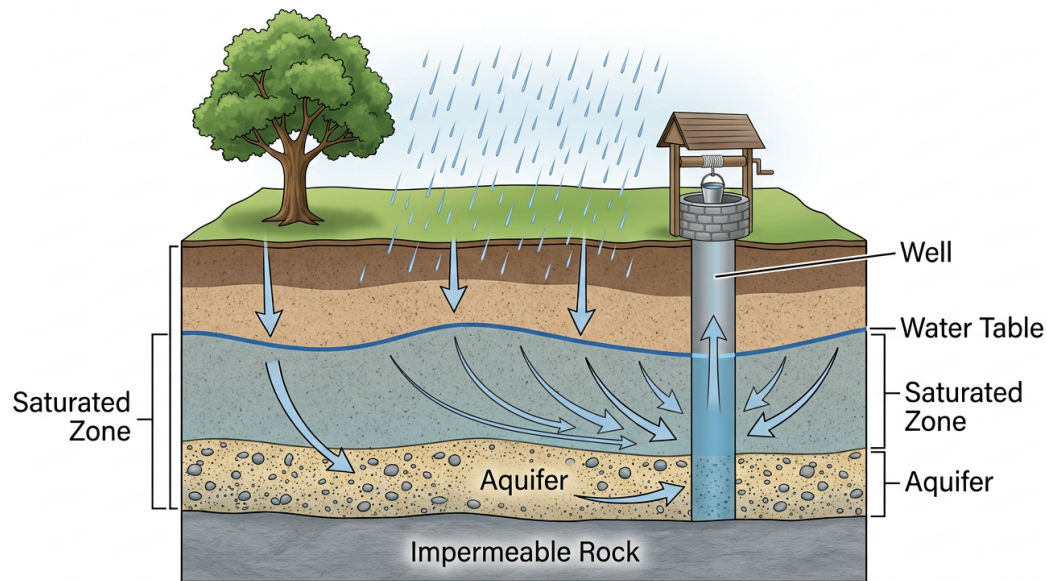
- Can be as **shallow as one foot** below the ground.
- Can also be **a few hundred metres deep**.
- **Heavy rains** → the water table **rises**.
- **Continuous extraction** of groundwater → the water table **falls**.

10. Groundwater Potential in India

India has **four major groundwater regions**.

10.1 Hard-Rock Aquifers of Peninsular India

- Represent around **65% of India's overall aquifer surface area**.
- Found in **central peninsular India**, underlain by hard-rock formations.
- Give rise to a **complex, extensive, low-storage** aquifer system.
- The **water level drops rapidly** once the water table falls by more than **2–6 metres**.
- Located **south of the Indo-Gangetic–Brahmaputra plains**.
- Rock types: **consolidated sedimentary rocks, Deccan Trap basalts, crystalline rocks**.



Groundwater, aquifers and the water table.

- States: **Karnataka, Maharashtra, Tamil Nadu, Andhra Pradesh, Orissa, Kerala.**
- Groundwater movement is **restricted to the weathered residuum and interconnected fractures** at deeper levels.
- **Limited groundwater potential** in this region.
- Rocks are weathered to a depth of about **30 m** under tropical conditions.
- Groundwater is found within a depth of **less than 50 m** (occasionally down to 100 m, rarely beyond).
- Development is largely through **dug wells**; **valley fills** are often dependable sources.
- Yield of wells in hard rock: **2–10 lps** (litres per second).

10.2 Alluvial Aquifers of the Indo-Gangetic Plains

- Found in the **Gangetic and Indus plains** of Northern India.
- Have **significant storage spaces** — a valuable source of fresh water.
- But **excessive extraction** and **low recharge rates** create a risk of **irreversible overexploitation**.
- The region covers about **850,000 sq km**.
- States: **Punjab, Haryana, Uttar Pradesh, Bihar, Assam, West Bengal.**
- Accounts for **more than 1/4th** of India's land area.
- Underlain by **thick piles of sediment** of Tertiary and Quaternary age.
- **Alluvial fill exceeds 1000 m** at places — the most productive groundwater reservoir in the country.
- Characterised by **regionally extensive, highly productive multi-aquifer systems**.

- Development is **still sub-optimal** in most states, except **Haryana and Punjab**.
- Deeper wells here yield **25–50 lps**.

10.3 Northern Mountainous Terrain and Hilly Areas

- Highly rugged mountainous terrain in the **Himalayan region**, extending from **Kashmir to Arunachal Pradesh**.
- Characterised by **steep slopes and high runoff**.
- Underlain by **granites, slate, sandstone, and limestone** (Palaeozoic to Cenozoic age).
- **Yield potential: 1 to 40 lps**.
- Offers **very little scope for groundwater storage**.
- However, acts as a **major source of recharge** for the vast Indo-Gangetic and Brahmaputra alluvial plains.

10.4 Coastal Areas

- Have a **thick cover of alluvial deposits** of Pleistocene to Recent age.
- Form **potential multi-aquifer systems** in **Gujarat, Kerala, Tamil Nadu, Andhra Pradesh, Orissa**.
- But **inherent quality problems** and the **risk of seawater ingress** severely constrain development.
- **Over-development** here raises the **risk of saline-water ingress**.
- Groundwater prospects vary from **5–25 lps** depending on local conditions.

Groundwater Availability in India (April 2015)

- Annual availability (natural runoff in rivers): **1,869 Billion Cubic Metres (BCM)/year**.
- Usable water resources: **1,123 BCM/year**.
- The difference is due to **constraints of topography** and the **uneven distribution** across river basins — so the entire 1,869 BCM/year **cannot be extracted**.

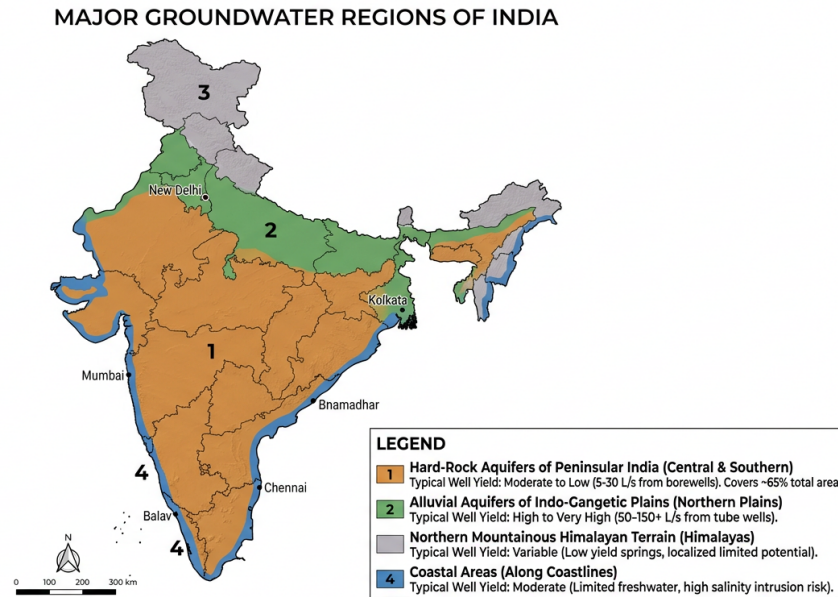
11. Conjunctive Use of Groundwater and Surface Water

Conjunctive Water Use

Conjunctive water use refers to the **simultaneous use of surface water and groundwater** to meet crop demand.

11.1 Explanation

- Each day, **hundreds of thousands of farmers** in canal, tank, and other surface-irrigation systems **combine surface water with groundwater**.



The four major groundwater regions of India.

- They do so in an **individual manner**, uncontrolled by any scheme or basin-level entity.

11.2 Optimising Conjunctive Use

- The best way forward is to **concentrate on capacity building** of irrigation systems.
- To sustain groundwater use in tube-well-irrigated areas, **enhancing recharge** from precipitation and surface-water imports is necessary.

12. Management of Groundwater

- Managing groundwater in the Indian context is an **extremely complex proposition**.
- It deals with the **interactions between human society and the physical environment**.
- There is a **highly uneven distribution** of groundwater availability and utilisation.
- **No single management strategy** can be adopted for the entire country.
- Each situation demands a **customised solution** that takes into account the:
 - Geomorphic set-up,
 - Climatic settings,
 - Hydrologic and hydrogeologic settings,
 - Groundwater availability,
 - Water-utilisation patterns across sectors, and
 - Socio-economic set-up of the region.

13. Groundwater Depletion

- **Indiscriminate groundwater development** has led to **substantial declines** in groundwater levels.
- The decline is seen in both **hard-rock and alluvial areas**, which threatens the **sustainability of the resource**.
- A **long-term decline** of groundwater levels is observed in many areas.

States Most Affected

Rajasthan, Gujarat, Tamil Nadu, Punjab, Delhi, and Haryana are the states most affected by groundwater decline.

13.1 Urban Water-Level Declines

- Seen in most cities that depend on groundwater for drinking water.
- **Water-level declines of up to 30 m or more** have been observed.

13.2 Traditional Water-Harvesting Methods

- Methods once in use in arid and semi-arid areas have either been **abandoned** or have **become defunct**.
- There is an **urgent need to revive these methods**.

14. Contamination of Groundwater

Groundwater Contamination

The **presence of certain pollutants in groundwater** that are in **excess of the limits prescribed for drinking water**.

14.1 Types of Contaminants

14.1.1 A. Geogenic Contaminants (Natural Origin)

Contaminant	Source
Arsenic	Natural rock leaching
Fluoride	Natural fluoride-rich minerals
Nitrate	Natural soil processes
Iron	Natural rock dissolution

14.1.2 B. Anthropogenic Contaminants (Human Activity)

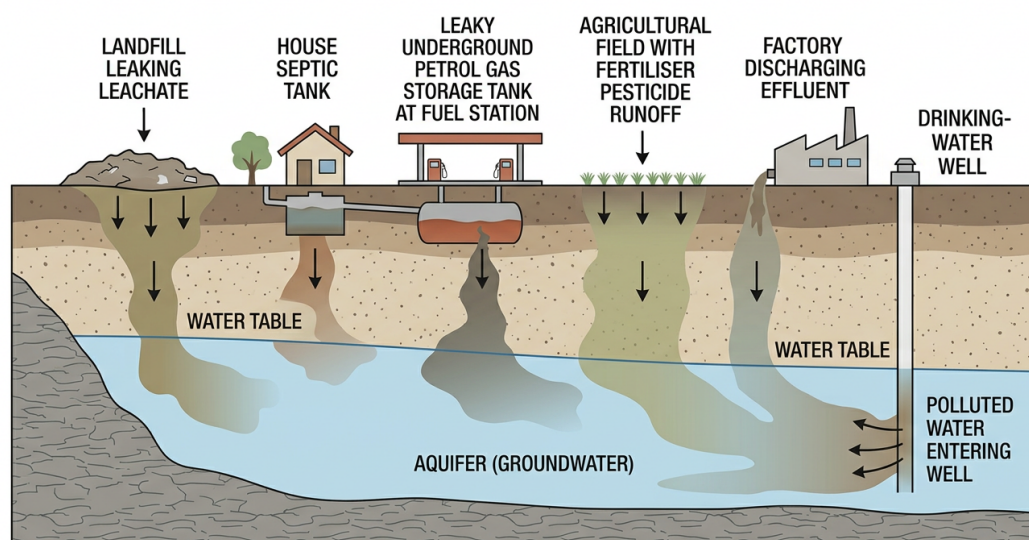
Contaminant	Source
Bacteria	Domestic sewage
Phosphates	Agricultural fertilisers
Heavy metals	Industrial effluents

14.2 Sources of Groundwater Contamination

- **Landfills** — leachate seeping into groundwater.
- **Septic tanks** — untreated sewage seeping underground.
- **Leaky underground gas tanks** — petroleum products contaminating aquifers.
- **Overuse of fertilisers** — nitrate and phosphate contamination.
- **Overuse of pesticides** — toxic chemical contamination.

Important Statistic

Nearly **60%** of all districts in India have issues related to either the **availability** of groundwater, or the **quality** of groundwater, or **both**.



Sources of groundwater contamination.

15. Recharge of Groundwater

15.1 Natural Recharge

- Flow below the land surface takes place through the process of **infiltration**.

- **Recharge of aquifers** happens during precipitation, as water infiltrates into the **pores in the soil strata**.
- The soil will **not become completely saturated** unless the water supply is maintained for **prolonged periods**.
- If water is applied **only intermittently**, there may be **no recharge** during the first infiltration, or even between two subsequent infiltrations.

15.2 Artificial Recharge of Groundwater

15.2.1 Why Artificial Recharge is Needed

- **Natural replenishment** of groundwater is a **slow process**.
- It often **cannot keep pace** with excessive, continued exploitation.
- This results in **declining groundwater levels** and **depletion** of resources.

Artificial Recharge

Artificial recharge aims at **augmenting the natural replenishment** of groundwater storage by some method of construction, by spreading of water, or by **artificially changing natural conditions**.

15.2.2 Uses of Artificial Recharge

- **Reducing overdraft** of groundwater.
- **Conserving surface run-off**.
- **Increasing available groundwater supplies**.

15.2.3 Types of Artificial Recharge

- **Incidental recharge** — a by-product of normal water-utilisation activities.
- **Deliberate recharge** — planned specifically to recharge groundwater.

15.2.4 Methods of Artificial Recharge

- **Planned methods** — storing water in **pits, tanks**, etc. to feed the aquifer.
- **Unplanned / incidental methods** — applied irrigation and leakages from pipes.

15.3 Artificial Recharge Techniques

15.3.1 A. Direct Techniques

i. Surface (spreading) methods:

1. **Flooding** — spreading water over large areas of land.
2. **Basins or percolation tanks** — water stored in basins/tanks to percolate into the ground.
3. **Stream augmentation** — adding water to streams to increase seepage.
4. **Ditch-and-furrow system** — shallow ditches and furrows to spread water.

ii. Sub-surface methods:

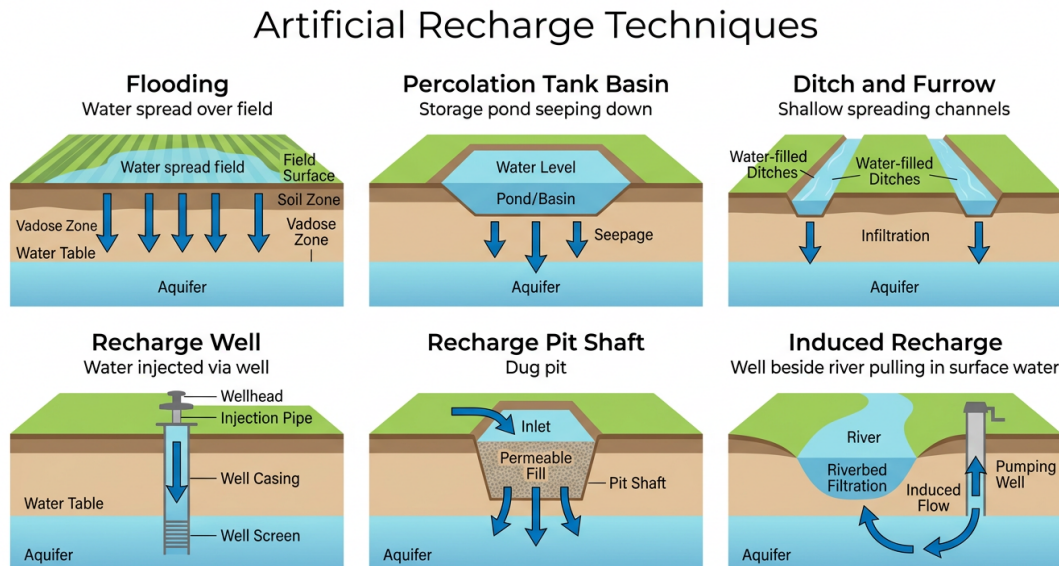
1. **Recharge well** — injecting water directly into the aquifer through wells.

2. **Recharge pit / shaft** — pits dug to allow water to seep into the ground.

3. **Dug well** — existing wells used for recharge.

15.3.2 B. Indirect Techniques

- **Induced recharge** — placing wells near rivers/streams to induce seepage from the surface-water body into the aquifer.



Methods of artificial groundwater recharge.

16. Seawater Ingress in Coastal Aquifers

16.1 Background

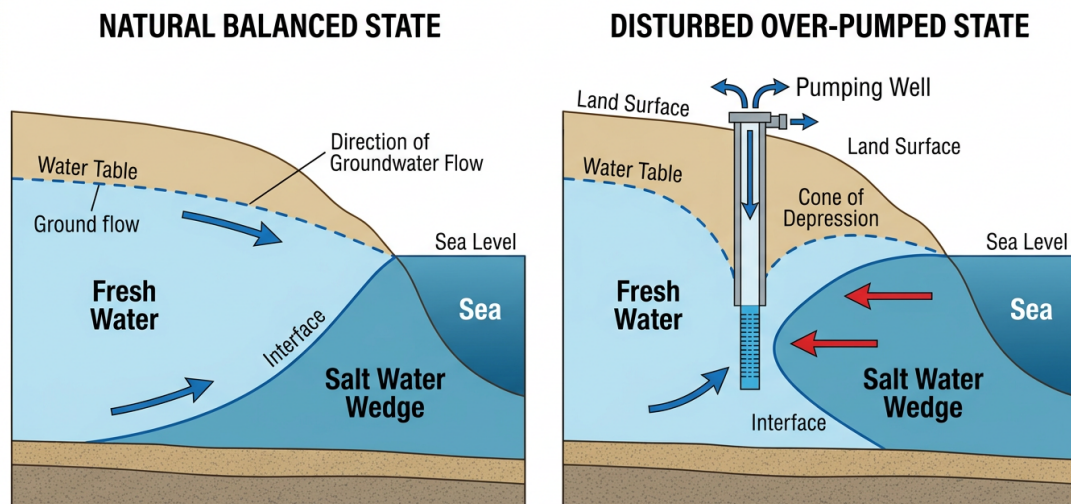
- Unconsolidated deltaic and coastal sediments form **thick, regionally extensive aquifers**.
- They have a **prolific yield potential** able to sustain deep, moderate-to-high-capacity tube wells.
- Considerable **fresh groundwater resources** have been identified along the east coast.
- However, **inherent quality problems** restrict their development.

16.2 What is Seawater Ingress (Intrusion)?

- Groundwater in coastal aquifers exists in a **fragile dynamic equilibrium with seawater**.
- **Indiscriminate exploitation** of groundwater from coastal aquifers **disturbs this equilibrium**.
- This results in the development of a **landward hydraulic gradient**.
- Which ultimately leads to **seawater intrusion into fresh-water aquifers**.

States Already Affected by Salinity Ingress

Gujarat, Tamil Nadu, and Andhra Pradesh.



Seawater intrusion into a coastal aquifer.

16.3 Measures to Prevent / Control Seawater Intrusion

Measure	Explanation
i. Regulate groundwater development in coastal areas	Controlled extraction to avoid over-pumping
ii. Form a freshwater ridge parallel to the coast (via artificial recharge)	Creating a barrier of fresh water to stop saltwater ingress
iii. Form a pumping trough using wells aligned parallel to the coast	A line of wells pumping out the seawater-intrusion zone
iv. Reduce water use	Conserving water reduces demand and hence extraction
v. Reduce pump depth	Shallow pumps extract less saline water than deep pumps
vi. Pump less water, more frequently	Small, frequent extractions are better than large volumes at once
vii. Locate wells at least 50 m from the coast	Distance from the coast reduces the risk of seawater mixing
viii. Avoid deep wells close to the coast	Deep coastal wells are the most vulnerable to saltwater intrusion

16.

Summary Table — Module 2: Water

Topic		Key Concept
Water Sources		Rain water, surface water, groundwater, reclaimed water
Global Water		97.5% saltwater; only 2.5% freshwater; most locked in glaciers
Indian Rivers		Indus, Ganga–Brahmaputra, peninsular rivers — drain to Arabian Sea and Bay of Bengal
Water Use Sectors		Agriculture (69%), Industry (15%), Domestic (15%), Recreation (small %)
Water Deficit in India		1/6th population but 1/25th water; $\sim 1600 \text{ m}^3/\text{capita}$ — under water stress
Inter-Basin Water Transfer		Transfer from surplus to deficit basins; aids flood/drought mitigation
Interlinking of Rivers		NWDA (1980); Himalayan (14 links) + Peninsular (16 links) components
Himalayan Component		Links Brahmaputra–Ganga–Mahanadi; 22 MHa irrigation + 30,000 MW
Peninsular Component		Network of peninsular basins; 13 MHa irrigation + 4000 MW
Merits of IBWT		Uniform water use, flood/drought control, employment, power, fisheries
Demerits of IBWT		Ecological damage, displacement, legal disputes, high cost, water loss
Groundwater		Water stored in aquifers; table rises with rain, falls with extraction
GW Potential in India		Hard-rock (65%), Indo-Gangetic alluvial, Himalayan terrain, coastal areas
Conjunctive Use		Simultaneous use of surface water + groundwater for crop demand
GW Depletion		Rajasthan, Gujarat, TN, Punjab, Delhi, Haryana most affected
GW Contamination		Arsenic, fluoride, nitrate, iron (natural); bacteria, heavy metals (human)
GW Recharge		Natural (infiltration) + artificial (flooding, percolation tanks, wells)
Seawater Ingress		Coastal over-pumping disturbs fresh–salt equilibrium → intrusion

Topic	Key Concept
Seawater Ingress Control	Regulate pumping, freshwater ridge, reduce pump depth, wells 50 m from coast

In a nutshell

These are the complete, detailed study notes of **Module 2: Water (BCV755B — Conservation of Natural Resources)**, covering every topic in the module — from the sources of water and global/Indian water resources, through inter-basin transfer and river interlinking, to groundwater, its recharge, contamination, and seawater ingress — without skipping anything.